MORGAN HUBERTY

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EDUCATION

Johns Hopkins University — Baltimore, MD
Bachelor of Science in Computer Engineering — *Expected May 2027*
Relevant Coursework: FPGA Lab, Computer System Fundamentals, Signals & Systems, Mastering Electronics, Computer Vision
Honors: Childhood Cancer Canada STEM Biomedical Survivor Scholarship (2024)

TECHNICAL SKILLS

Hardware Design: Digital design, RTL (VHDL, Verilog); FSMs, CDC (two-flop synchronizers), UART, I²C & SPI, open-drain I/O
Verification & EDA: Self-checking testbenches, assertions, Verilator (C++ drivers), Xilinx Vivado (synthesis, place and route)
Lab & Analysis: Oscilloscope, multimeter, LTSpice; circuit analysis; analog and digital signals (continuous- and discrete-time)
Software: C, C++, Python, Bash, Git, Linux; Arduino firmware; basic x86-64 assembly; GenAI tools (Claude, Gemini, Cursor)
Currently Learning: SystemVerilog, SystemC | **Interests:** Low-power ASIC and SoC design for edge AI and LLM inference

HARDWARE DESIGN PROJECTS

ZipCPU Soft-Core Processor Build | *Verilog, Verilator, C++, Artix-7* — In Progress
- Working through Dan Gisselquist's ZipCPU tutorial curriculum toward building the ZipCPU, a 32-bit pipelined RISC soft-core processor, on a Digilent Cmod A7-35T (Xilinx Artix-7)
- Writing Verilog RTL and simulating each design with Verilator and C++ test drivers before synthesizing and validating it on hardware; formal verification with SymbiYosys is the next milestone

UART Transceiver | *VHDL, Xilinx Vivado, Cmod A7-35T (Artix-7)* — github.com/mhubert3/FPGA-Serial-UART
- Designed a parameterized UART transmitter and receiver (8N1, 115,200 baud from a 12 MHz clock) with FSM control, a two-flop synchronizer on the asynchronous RX input, and mid-bit sampling via a 1.5-bit-period start delay
- Verified with a self-checking closed-loop testbench using VHDL assertions on a worst-case alternating pattern, then validated on hardware through PMOD loopback and a MATLAB serial host interface

I²C Master Controller | *VHDL, Xilinx Vivado, Cmod A7-35T (Artix-7)* — github.com/mhubert3/FPGA-Serial-I2C
- Designed an FSM-based I²C master handling start, repeated start, stop, read/write, ACK/NACK, and clock stretching, with cycle-counted 5 µs bus timing and Xilinx IOBUF primitives for open-drain SDA/SCL
- Modeled bus pull-ups as weak-high signals in simulation; validated end-to-end on hardware by relaying bytes from a MATLAB GUI over UART to the FPGA, across I²C to a PIC16F18326 slave, and back

EXPERIENCE

Morgate Solutions | *Software Engineering Intern* — Summers 2024, 2026
Edge Software Optimization — *Jun 2026 – Aug 2026*
- Containerized C/C++ parsers with Docker for Linux edge gateways, optimizing memory use to fit 512 MB device limits
- Emulated network jitter and profiled CPU usage to find and prevent buffer overflows in high-frequency data streams

Low-Level Data & Protocol Bridging — *Jun 2024 – Aug 2024*
- Developed C extraction tools using struct packing and bitwise operations to decode 50+ GB of binary telemetry
- Managed struct byte alignment and endianness; built a Python test harness to validate parser output

COMPUTER ARCHITECTURE PROJECT

Cache Simulator | *C, C++* — github.com/mhubert3/Cache-Simulator
- Built a configurable set-associative cache simulator (sets, blocks, block size, LRU/FIFO replacement, write policies) that reports hit/miss counts and cycle totals from memory traces, used to compare associativity and policy trade-offs

LEADERSHIP

National Society of Black Engineers, JHU Chapter | *Co-President; former Vice President & Finance Chair* — Sep 2023 – Present
- Secured sponsorships from Medtronic, Visa, and Jane Street; hosted events with Medtronic and Shift Alt Her
- Grew reach via Life Design Lab and Pava Center events and collaborations with five student orgs, including SHPE and BMES
- Building the chapter website; grew outreach on Instagram and LinkedIn and started alumni LinkedIn and WhatsApp groups
- As Finance Chair, managed a $9K annual budget and secured $3.5K through fundraising and corporate sponsorships

HopThon, JHU Dance Marathon | *Fundraising Chair* — Fall 2025 – Spring 2026
Alpha Epsilon Pi | *Philanthropy Chair* — Spring 2025